AYAM BANJADE

Github Linkedin Portfolio

Alte Hafenstraße 9,
28757 Bremen

ayam.banjade@gmail.com

+4915752730515

Nepalese

30-11-2004

ADDITIONAL SKILLS

Rust | React | Flutter | TypeScript | CMake | Linux | Command-line Interface | XML | Agile Development | Shell Script | Machine Learning | Docker

LANGUAGES

English:
Native / Bilingual (C2)

German:
Limited working proficiency (A2)

Hindi:
Business fluent

Nepali:
Native / Bilingual (C2)

EDUCATION

Constructor University Bremen — **Sept. 2021 - Jun. 2024 (Estd.)**
B.Sc. Computer Science, Mathematics minor — Bremen, Germany

- Courses: Software Engineering, Databases and Web Services, Algorithms and Data Structures, Computer Networks, Operating Systems, Distributed Algorithms, Secure and Dependable Systems

EXPERIENCE

Constructor University Bremen - Teaching Assistant — **Bremen, Germany**

Numerical Methods — Feb. 2023 - Jun. 2023

- Created and solved challenging problems and helped 100+ students with numerical methods for root finding, interpolation, approximation, numerical differentiation / integration / differential equations, and their implementations in **Python**
- Streamlined the grading process by developing Python scripts transforming and simplifying complex CSV files

Programming in C/C++ — Sept. 2022 - Dec. 2022

- Debugged and validated **C/C++** code, providing comprehensive tutorials to 200+ students on programming concepts, including pointers, object-oriented programming, dynamic memory allocation, data structures, templates, macros, header files, and makefiles

Introduction to Computer Science — Sept. 2022 - Dec. 2022

- Coordinated and led tutorials addressing students' inquiries in functional programming (**Haskell**), assembly programming, computer science foundations, discrete mathematics, Boolean algebra, Hoare's logic, and digital circuit design

PROJECTS

Rippit

- Developed a frontend and backend of a forum website collaboratively using **VCS (git)** and integrated search suggestions and map functionality using jQuery and Leaflet library
- Implemented backend using **PHP** and **MySQL** to connect the website and **MongoDB** database server with query-dependent access rights using sanitized SQL queries

Bug World

- Collaborated in agile sprints to build and improve different code bases for a full-stack simulation game in **JavaScript** and **HTML/CSS**
- Engineered and implemented a parser, assembler, simulator, and GUI using OOP principles and created automated test cases to ensure robustness and functionality
- Authored and maintained detailed technical documentation and comprehensive specifications to ensure effective project communication and development

Room Booking System

- Designed and built a **Java**-based, user-friendly, and intuitive CLI room booking system with file I/O, dynamic matching, and custom data structures for efficient data management

Spam Email Extracter

- Built a C++ program to implement data extraction techniques to retrieve 57 relevant attributes from UCI's Spambase Data Set for effective spam email detection

Stochastic Methods Lab

- Developed stochastic models for financial mathematics using Python libraries (`matplotlib`, `NumPy`, `pandas`, `SciPy`) for CSV handling and complex dataset analysis and visualization

Mars Photo App

- Created a **Kotlin** app displaying Mars' surface photos taken by NASA rovers using Retrofit for REST requests, Moshi for JSON deserialization, and Coil for image loading via URLs
- Leveraged coroutines, ViewModel, LiveData, and Data Binding with binding adapters to enhance performance and functionality

Bremen, Germany, 20/06/2023 Ayam